the RIKZ (Berrevoets et al., 2003). Older counts, which are available from 1978 onwards, have been analyzed as part of EVA II, after correcting for missing counts through imputing (Rappoldt et al., 2003b). Compared to the Delta area, the frequency of counts in the Wadden Sea covering the entire area has been decidedly lower, on average three times a year (Meltofte et al., 1994; van Roomen et al., 2003). The longest uninterrupted counting series for the Wadden Sea (from 1973/1974 onwards) is available for the month of January. A logical choice is therefore to restrict analyses of bird numbers to this month (Leopold et al., 2003b; Verhulst et al., 2004). However, January counts have a problem due to ice winters. When the mud flats become covered with ice during a sufficiently long period of frost, a large number of oystercatchers may leave the Wadden Sea to winter elsewhere (Hulscher, 1989; Hulscher et al., 1996; Camphuysen et al., 1996). During the past century, such hard-weather movements from the Dutch Wadden Sea have occurred in at least 10 of 92 years (Hulscher et al., 1996). Very low counts of oystercatchers in the winters of 1986/87 and 1996/97 are due to hard-weather movement and not to a change in the number of oystercatchers that arrived in autumn to spend the winter in the area. One solution is to include integral counts for other months in the imputing exercise. To the 29 integral counts for January, 33 counts during other months in the period August-December can be included (Leopold et al., 2003b). This provides only a partial solution. For the winter of 1996/97 there were additional counts in August and December, so the new estimate is clearly higher. However, for the winter of 1986/87 only the January count is available, so the procedure makes no difference. Apart from integral counts of the whole Dutch Wadden Sea, there are many sites throughout the Wadden Sea that are counted on a much more regular basis. These counts can also be included in the imputing to yield an estimate of the number of oystercatchers in each site in each month. For the studies of Leopold et al. (2003b) and Rappoldt et al. (2003a) counting data of individual counting sites were first clustered into larger areas and then imputing was done for these larger cluster areas. More recently, SOVON has performed imputing directly on the individual counting sites, without first clustering these sites into larger areas (van Roomen & van Winden, pers. comm.). Including the counts of sites counted more regularly leads to estimated numbers that vary smoothly over the years. Such gradual changes in numbers are what one would expect for a long-lived species like the oystercatcher. We therefore conclude that imputing using all counts offers the best estimate of the number of oystercatchers deciding to spend the winter in the Dutch Wadden Sea. The precise estimate of the total number depends on the details of the imputing and this requires further study. A distinction was made between areas open for fishing, areas closed for shellfish fishing and areas that could not be classified (Figure 7). Since bird counts take place during high tide, the assignment to the low tide feeding areas had to be based on expert judgement and is open to further improvement. Especially for species that travel long distances to reach the feeding grounds, errors are likely. 